

Refugee Inflows and Native Parity Progression

Evidence from the Syrian inflow and Turkish DHS

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Why this question, why Turkey.

- Global fertility is below replacement ($\text{TFR} < 2.1$) in nearly all high- and middle-income countries; raising native fertility is now an explicit policy goal in many.
- Turkey's TFR **fell from 2.10 in 2010 to ≈ 1.62 in 2022** — below replacement, below the OECD average — while at the same time becoming the **largest refugee host** ($\approx 3.6\text{M}$ Syrians, post-2011).
- Turkey is therefore a uniquely sharp setting to study the **interaction between low native fertility and a large migration shock**.

Question. How did the post-2011 inflow affect native Turkish women's marriage and fertility outcomes **by parity and by age**?

Why parity-by-age. Two competing mechanisms make distinct predictions:

- **Marriage-market crowd-out** $\Rightarrow$ delays *first births*, depresses *marriage*, effects in early 20s.
- **Opportunity-cost / household-production** $\Rightarrow$ lowers cost of additional births where childcare/domestic labour is cheaper, effects at *higher parities*, *late 20s–30s*.

A pooled total-fertility regression cannot distinguish these. **The parity-age grid can.**

Wave	Sample	Controls	Province assignment
2003	Ever-married only	Limited	Childhood province only
2008	Ever-married only	Limited	Childhood province only
2013	Ever-married <i>and</i> never-married	Full	2011 residence (mig. history)
2018	Ever-married <i>and</i> never-married	Full	2011 residence (mig. history)

Two analytical samples:

- **Primary** — 2013/18, full sample (incl. never-married), province in 2011. Most accurate pre-shock province assignment, no pre-selection on marriage.
- **Four-wave** — 2003/08/13/18, ever-married only, childhood province. Anchors trend estimation with pre-shock cohorts; restricted by data availability in 2003/08.

Today's results. Headline pooled results presented on the primary sample; four-wave estimates and decomposition ladder in the appendix.

Cumulative parity outcomes (absorbing-state indicators):

$$Y_{ija}^{(p)} = \mathbb{1}\{\text{parity}_i \geq p \text{ by age } a\}, \quad p \in \{1, 2, 3, 4, 5\}, \quad a \in \{16, 20, 24, 28, 32, 36\}.$$

Plus marriage analogue $\mathbb{1}\{\text{ever married by age } a\}$.

Exposure: average refugee-to-native ratio over the woman's age window from 12 to a :

$$\bar{R}_{ja} = \frac{1}{a - 11} \sum_{s=12}^a R_{j, t(s,i)}.$$

Interpretation. A 1 pp. higher *average annual* ratio over ages 12– a raises $\Pr(\text{parity} \geq p \text{ by } a)$ by $\hat{\beta}^{(p,a)}$ pp.

Cohort dependence. Two women aged a in the same province in different waves get different $\bar{R}_{ja}$ — their averaging windows cover different calendar years. This cross-cohort variation drives identification.

Province-year instrument (Aksu, Erzan and Kirdar 2022):

$$Z_{n,t} = \sum_{s=1}^{13} \underbrace{\left[\frac{(1/d_{s,T}) \pi_s}{\sum_{X \in \{T,L,J,I\}} (1/d_{s,X})} \right]}_{\text{share weight (time-invariant)}} \cdot \frac{T_t}{d_{n,s}}$$

π_s pre-war pop. share of Syrian province s ; $d_{s,X}$ distance from s to host $X \in \{T, L, J, I\}$; $d_{n,s}$ distance from s to Turkish province n ; T_t aggregate Syrian refugee stock across the four host countries.

Key decomposition — only T_t varies in time:

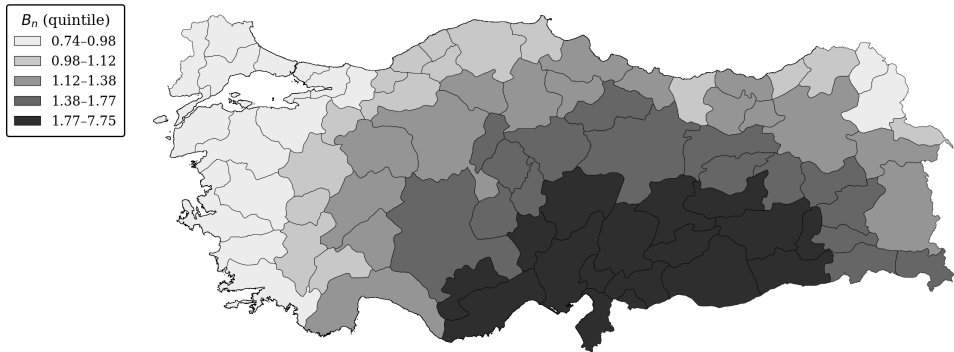
$$Z_{n,t} = B_n \times T_t, \quad \bar{Z}_{ja} = B_j \times \bar{T}_a.$$

$\implies$ **Continuous-treatment difference-in-differences**: B_n assigns province intensity (geography), T_t activates the shock after 2011.

Exclusion restriction collapses to one condition: B_n uncorrelated with province-specific *pre-existing trends* in fertility, conditional on FE and trend controls.

Settlement-Propensity Weight B_n

Settlement-propensity weight B_n across Turkish provinces (quintiles)



Estimating Equation

For each (p, a) separately,

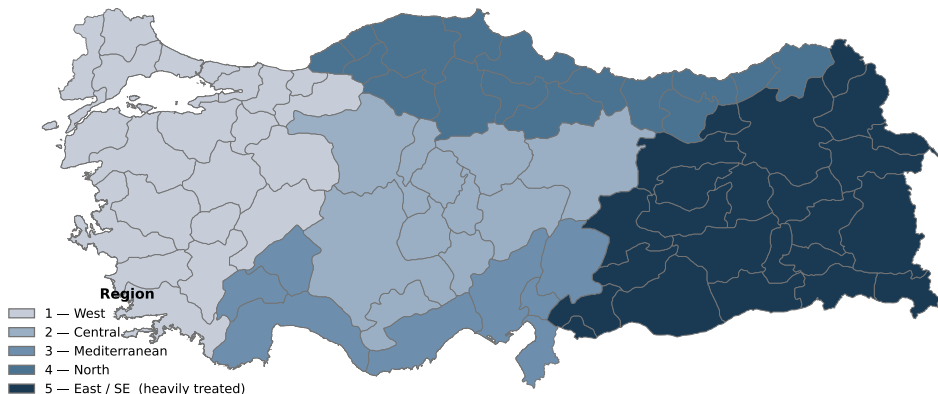
$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)} \bar{R}_{ja} + X_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} \cdot t(a, i) + \lambda_{l(i)} \cdot t(a, i) + \varepsilon_{ija},$$

with $\bar{R}_{ja}$ instrumented by $\bar{Z}_{ja} = B_j \times \bar{T}_a$.

- δ_j — province FE $\mu_{t(a,i)}$ — year-at-age- a FE ($\equiv$ birth-cohort FE at fixed a)
- $\theta_{g(j)} \cdot t$ — **five-region**-specific linear time trends
- $\lambda_{l(i)} \cdot t$ — **mother-tongue**-specific linear time trends (Turkish / Kurdish / Arabic)
- Trends start in **1987**; X_i : wave, childhood urban/rural, mother's education
- Standard errors clustered at province, DHS sampling weights applied

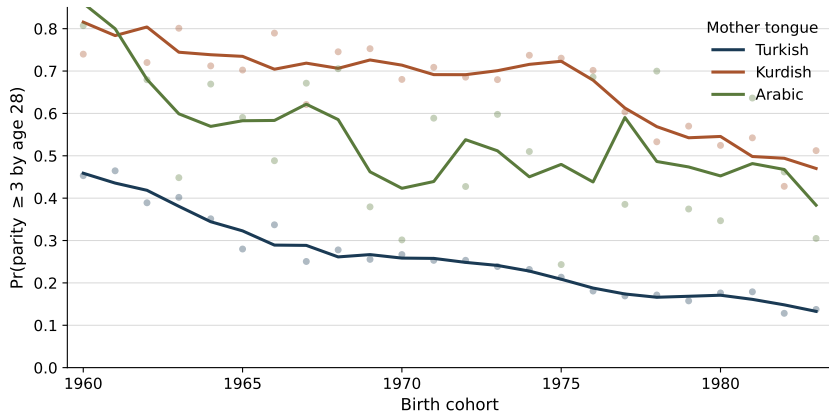
We motivate the two trend controls (region and mother tongue) on the *next two slides*: each absorbs a pre-existing differential trend that is correlated with B_n by geography or ethnicity.

Why Region-Specific Trends? — The 5-Region Scheme



Region 5 (East/SE) houses $\approx 95\%$ of refugees, has **higher baseline fertility**, and was on a **distinct demographic transition** relative to the rest of the country pre-2011 — without region-specific trends, $B_n \times T_t$ would absorb these dynamics. Five regions chosen for parsimony (NUTS-1 / NUTS-2 in appendix).

Why Mother-Tongue-Specific Trends?



Pre-shock cohorts only. **Levels and trends differ sharply** across MT groups — and Region 5 (high B_n) is **55% Kurdish**-speaking versus 1.7–12% in regions 1–4. So B_n correlates with MT composition by geography, and MT-specific trends are needed to absorb the differential transition.

Placebo: $Y \sim B_n \times t$ on Pre-Shock Cohorts (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003 (0.0005)	0.0010 (0.0007)	0.0012 (0.0007)	0.0008 (0.0004)	0.0011 (0.0006)	-0.0001 (0.0007)
Parity ≥ 1	0.0005 (0.0003)	0.0011 (0.0008)	0.0007 (0.0010)	0.0011 (0.0005)	0.0011 (0.0005)	0.0001 (0.0007)
Parity ≥ 2	0.0001 (0.0002)	-0.0001 (0.0004)	0.0009 (0.0006)	0.0018 (0.0007)	0.0015 (0.0010)	0.0006 (0.0011)
Parity ≥ 3	0.0000 (0.0000)	-0.0000 (0.0002)	0.0004 (0.0007)	0.0000 (0.0008)	-0.0007 (0.0008)	-0.0024 (0.0013)
Parity ≥ 4	– (–)	– (–)	0.0002 (0.0004)	0.0011 (0.0009)	0.0005 (0.0023)	-0.0005 (0.0022)
Parity ≥ 5	– (–)	– (–)	0.0000 (0.0002)	0.0002 (0.0005)	0.0012 (0.0014)	0.0018 (0.0022)
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Test. On the subsample $\{i : t(a, i) < 2012\}$, regress each outcome on $B_n \times t(a, i)$ with the full set of FE and trends. After Romano–Wolf stepdown across the 32-cell family, **no cell rejects**. First-stage $F \in [43, 144]$ (FS); raw + four-wave [appendix](#).

Main Results — Primary Sample (raw analytical p)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0467 (0.2165)	-0.5715 (0.6000)	0.5130 (1.6892)	-0.0130 (1.0473)	0.2202 (0.9200)	-0.2993 (1.2412)
Parity ≥ 1	0.0095 (0.1273)	0.1226 (0.5231)	-0.0431 (1.9677)	0.9235 (1.3419)	-1.5150 (1.7893)	0.1839 (1.3711)
Parity ≥ 2	0.0285 (0.0409)	0.3048 (0.5464)	-0.1508 (0.7457)	0.6126 (1.9059)	1.7403 (3.4194)	-0.8450 (1.1891)
Parity ≥ 3	0.0084 (0.0055)	0.3816 (0.2481)	-0.3712 (0.5649)	1.2502 (1.5588)	2.8929 (2.6999)	1.0620 (1.8097)
Parity ≥ 4	— (—)	— (—)	0.2496 (0.3399)	1.9047** (0.8094)	2.2458 (1.4146)	4.2520** (1.9632)
Parity ≥ 5	— (—)	— (—)	0.3133 (0.2324)	1.3921* (0.8025)	1.4136 (0.9119)	4.1326** (1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Pooled **positive at higher parities, late 20s/30s** (raw stars). Pooled-primary RW in progress; under BH (BH) no cell survives, so read as *suggestive*. **Signal sharp in heterogeneity** (next two). Marriage / parities 1–3 null.

Four-wave (EM)

Heterogeneity I — Heavily-Treated Region (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2452 (0.3582)	-1.6915 (0.6464)	-2.5905 (1.9762)	-0.9127 (1.6556)	-0.4301 (0.7606)	1.5648 (0.8548)
Parity ≥ 1	-0.0233 (0.1541)	-0.4039 (0.4780)	-3.0301 (1.5690)	1.4459 (2.2608)	0.3391 (1.4282)	1.9888 (1.1851)
Parity ≥ 2	0.0210 (0.0518)	1.4250 (0.4706)	1.2829 (0.9335)	0.4696 (2.1259)	5.4953 (2.4693)	0.3303 (1.1054)
Parity ≥ 3	-0.0151 (0.0193)	0.3390 (0.4285)	-0.0495 (0.5988)	5.1117 (2.0445)	9.1213** (2.8160)	4.4002 (1.7627)
Parity ≥ 4	— (—)	— (—)	0.2292 (0.5746)	2.9874 (1.2826)	6.7512** (1.6983)	5.7527 (2.4905)
Parity ≥ 5	— (—)	— (—)	-0.0136 (0.4266)	1.8753 (0.9337)	4.1483 (1.3901)	6.5940 (3.4988)
Observations	4,002	3,522	3,022	2,404	1,851	1,311

Region 5 under Romano–Wolf: **parity ≥ 3 @ 32 = +9.12 pp (**)** and **parity ≥ 4 @ 32 = +6.75 pp (**)**.
 Region 1–4 entirely null (**region 1–4**).

Heterogeneity II — Mother Tongue (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.5562 (0.3233)	0.5199 (1.7594)	0.5854 (2.4777)	-4.0251 (2.2391)	1.9332 (2.2384)	0.8593 (1.1457)
Parity ≥ 1	0.3356 (0.1769)	1.8205 (1.5930)	-1.7541 (2.3234)	-1.5852 (2.3963)	2.4411 (2.7699)	0.7623 (1.8112)
Parity ≥ 2	0.0471 (0.0729)	1.8055 (1.1665)	1.4326 (1.8375)	1.4601 (2.9870)	4.4175 (3.2879)	1.9937 (1.4012)
Parity ≥ 3	-0.0192 (0.0258)	0.6474 (1.2286)	-0.8687 (1.2296)	12.3048** (2.4693)	6.0191 (3.4792)	2.5694 (1.5223)
Parity ≥ 4	— (—)	— (—)	1.2013 (1.3175)	8.1188 (1.7515)	2.0236 (3.8106)	8.2440* (1.7884)
Parity ≥ 5	— (—)	— (—)	0.3944 (1.0175)	3.1548 (2.4496)	4.7186 (3.3471)	5.0376 (2.3657)
Observations	3,145	2,785	2,415	1,951	1,489	1,055

Kurdish/Arabic, RW: **parity ≥ 3 @ 28 = +12.30 (**)**, **parity ≥ 4 @ 36 = +8.24 (*)**. Turkish null (**Turkish**). Low-educ, village (**low educ** **village**) match in point estimates, lose stars. **Common thread: higher-baseline-fertility** groups — room to move up.

Decomposition: Where in the Parity Ladder?

Discrete-time hazard models, current-year exposure $R_{j(i,t),t}$.

For each transition k :

$$h_{ijt}^{(k)} = \alpha + \beta^{(k)} R_{j(i,t),t} + \mathbf{X}_i' \Gamma + \phi_{\text{age}}^{(k)} + \mathbb{1}\{k \geq 2\} \psi_{\tau}^{(k)} + \delta_j + \mu_t + \theta_{g(j)} t + \lambda_{l(i)} t + \varepsilon_{ijt}^{(k)}.$$

Transition	Marriage	1st birth	2nd birth	3rd birth	4th birth	5th birth
$R_{j(i,t),t}$	0.0396 (0.0587)	-0.0502 (0.0442)	-0.0336 (0.1876)	0.0698 (0.0784)	-0.0024 (0.0661)	0.0162 (0.0646)
Person-yrs (N)	—	—	—	—	—	—

Reading. Pooled hazards are **imprecise zeros** (all $p_{RW} > 0.74$). Two non-contradictory reasons: (i) the hazard model *averages over years and women*; the parity-by-age pooled primary results in the main table are also null — the signal lives in subgroups, where subgroup hazards are work in progress; (ii) hazards are *conditional* on reaching the prior parity, so they answer "does exposure shift the 4th-birth hazard given 3 births already?" — a different question from "does exposure raise the share ever at parity ≥ 4 " estimated by the Stage 3 absorbing-state outcomes.

Cross-sectional 2SLS, four-wave EM sample, current-year exposure.

Outcome: $\mathbb{1}\{\text{ideal number of children} \geq k\}$ for $k \in \{2, 3, 4, 5\}$.

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0458 (0.0605)	0.1739 (0.2536)	0.5142 (0.2700)	0.2555 (0.1087)
Observations	14,775	14,775	14,775	14,775

Romano–Wolf adjusted across the 4-test family. Raw point estimates: $\Pr(\text{ideal} \geq 5)$ shifts by +0.26 pp per pp of average ratio (raw $p < 0.05$); ≥ 4 by +0.51 (raw $p < 0.10$); $\geq 2, \geq 3$ flat. Under RW $\{0.68, 0.68, 0.27, 0.21\}$, **nothing survives correction in the pooled EM sample**, but the **point-estimate pattern aligns with realised fertility**: preferences shift up at the same parities. **Subgroup ideal- n does survive RW for Kurdish/Arabic and Region 5** — see [appendix](#).

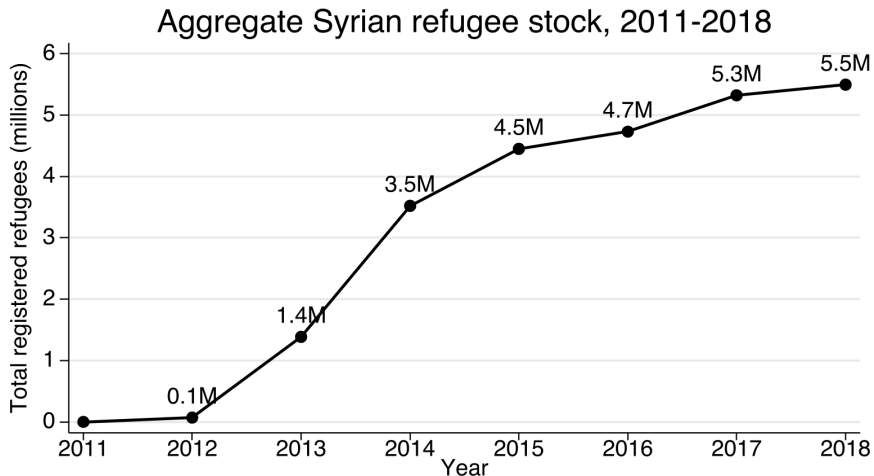
The Syrian inflow raised native parity progression — in groups with higher baseline fertility.

- Pooled primary positive at parities 4–5, ages 28/36 (raw stars); pooled-primary RW correction in progress.
- Under **Romano–Wolf** stepdown:
 - **Region 5**: parity ≥ 3 @ 32 = +9.12 pp (**); parity ≥ 4 @ 32 = +6.75 pp (**).
 - **Kurdish/Arabic**: parity ≥ 3 @ 28 = +12.3 pp (**); parity ≥ 4 @ 36 = +8.24 pp (*).
- Marriage and first births **unaffected** everywhere — rules out marriage-market crowd-out as the dominant mechanism.
- Subjective **ideal- n** : ≥ 4 , ≥ 5 shift up in the pooled (raw); RW-significant in Region 5 and Kurdish/Arabic subgroups.

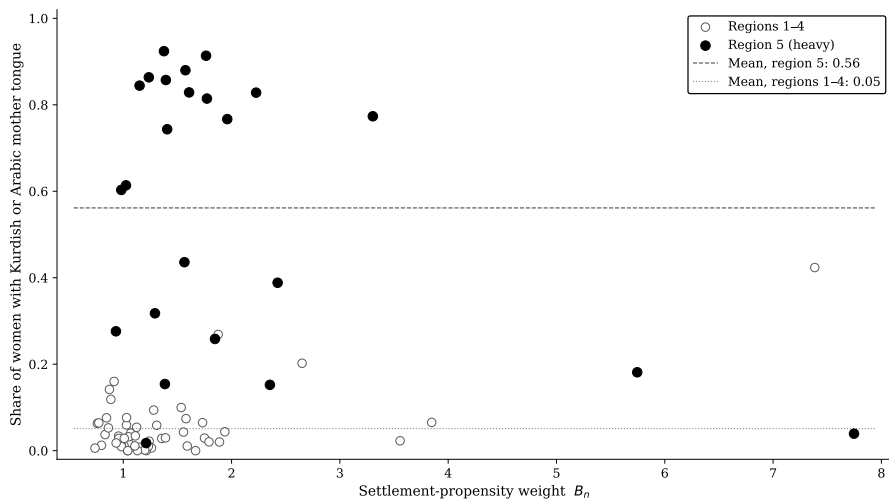
Mechanism. Consistent with an **opportunity-cost / household-production** channel: refugee inflows lowered the price of household services and informal childcare, easing additional births — response concentrated where there is room to move up the parity ladder.

Thank you. / Questions?

Appendix



Appendix: B_n vs Kurdish/Arabic Share at Province Level



Each dot is one province. Provinces with high B_n are also those with high Kurdish/Arabic shares — motivating the MT-specific trend control.

Appendix: Main IV — Primary Sample (BH-FDR adjusted)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0467 (0.2165)	-0.5715 (0.6000)	0.5130 (1.6892)	-0.0130 (1.0473)	0.2202 (0.9200)	-0.2993 (1.2412)
Parity ≥ 1	0.0095 (0.1273)	0.1226 (0.5231)	-0.0431 (1.9677)	0.9235 (1.3419)	-1.5150 (1.7893)	0.1839 (1.3711)
Parity ≥ 2	0.0285 (0.0409)	0.3048 (0.5464)	-0.1508 (0.7457)	0.6126 (1.9059)	1.7403 (3.4194)	-0.8450 (1.1891)
Parity ≥ 3	0.0084 (0.0055)	0.3816 (0.2481)	-0.3712 (0.5649)	1.2502 (1.5588)	2.8929 (2.6999)	1.0620 (1.8097)
Parity ≥ 4	— (—)	— (—)	0.2496 (0.3399)	1.9047 (0.8094)	2.2458 (1.4146)	4.2520 (1.9632)
Parity ≥ 5	— (—)	— (—)	0.3133 (0.2324)	1.3921 (0.8025)	1.4136 (0.9119)	4.1326 (1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Appendix: Main IV — Four-Wave (EM), BH-Adjusted

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.3132 (1.9142)	-1.3481 (1.2286)	0.9856 (2.2382)	-0.6077 (1.1088)	-1.3975 (1.4367)	1.3735 (0.6797)
Parity ≥ 2	0.3222 (0.1720)	1.2727 (2.2397)	0.6867 (1.4517)	-1.0745 (2.0183)	2.2526 (2.6152)	0.0998 (1.9274)
Parity ≥ 3	0.1429 (0.1168)	1.4820 (0.9743)	-0.7720 (0.9236)	-0.9003 (1.9910)	0.7142 (2.3653)	-0.5607 (2.5075)
Parity ≥ 4	— (—)	— (—)	-0.3685 (0.5178)	0.4007 (0.7539)	0.0293 (1.5816)	2.7419 (2.0043)
Parity ≥ 5	— (—)	— (—)	0.2797 (0.3048)	0.7892 (0.5506)	-0.3457 (1.2736)	1.5672 (1.6244)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

[← Back](#)

Appendix: Main IV — Four-Wave (EM), Raw Stars

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.3132 (1.9142)	-1.3481 (1.2286)	0.9856 (2.2382)	-0.6077 (1.1088)	-1.3975 (1.4367)	1.3735** (0.6797)
Parity ≥ 2	0.3222* (0.1720)	1.2727 (2.2397)	0.6867 (1.4517)	-1.0745 (2.0183)	2.2526 (2.6152)	0.0998 (1.9274)
Parity ≥ 3	0.1429 (0.1168)	1.4820 (0.9743)	-0.7720 (0.9236)	-0.9003 (1.9910)	0.7142 (2.3653)	-0.5607 (2.5075)
Parity ≥ 4	— (—)	— (—)	-0.3685 (0.5178)	0.4007 (0.7539)	0.0293 (1.5816)	2.7419 (2.0043)
Parity ≥ 5	— (—)	— (—)	0.2797 (0.3048)	0.7892 (0.5506)	-0.3457 (1.2736)	1.5672 (1.6244)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Appendix: Placebo — Primary Sample (Raw)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003 (0.0005)	0.0010 (0.0007)	0.0012* (0.0007)	0.0008** (0.0004)	0.0011* (0.0006)	-0.0001 (0.0007)
Parity ≥ 1	0.0005* (0.0003)	0.0011 (0.0008)	0.0007 (0.0010)	0.0011** (0.0005)	0.0011* (0.0005)	0.0001 (0.0007)
Parity ≥ 2	0.0001 (0.0002)	-0.0001 (0.0004)	0.0009 (0.0006)	0.0018** (0.0007)	0.0015 (0.0010)	0.0006 (0.0012)
Parity ≥ 3	0.0000 (0.0000)	-0.0000 (0.0003)	0.0004 (0.0007)	0.0000 (0.0009)	-0.0007 (0.0008)	-0.0024* (0.0014)
Parity ≥ 4	— (—)	— (—)	0.0002 (0.0004)	0.0011 (0.0009)	0.0005 (0.0023)	-0.0005 (0.0022)
Parity ≥ 5	— (—)	— (—)	0.0000 (0.0002)	0.0002 (0.0005)	0.0012 (0.0014)	0.0018 (0.0023)
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Appendix: Placebo — Four-Wave (EM), BH

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.0003 (0.0002)	0.0004 (0.0005)	0.0003 (0.0005)	0.0008 (0.0004)	0.0006 (0.0002)	0.0002 (0.0002)
Parity ≥ 2	0.0001 (0.0001)	0.0002 (0.0003)	0.0004 (0.0003)	0.0003 (0.0005)	0.0014 (0.0005)	0.0005 (0.0003)
Parity ≥ 3	0.0000 (0.0000)	0.0002 (0.0001)	-0.0002 (0.0004)	-0.0009 (0.0004)	-0.0007 (0.0003)	-0.0008 (0.0006)
Parity ≥ 4	— (—)	— (—)	-0.0004 (0.0002)	-0.0011 (0.0004)	-0.0013 (0.0008)	-0.0016 (0.0009)
Parity ≥ 5	— (—)	— (—)	0.0002 (0.0001)	-0.0009 (0.0003)	-0.0013 (0.0008)	-0.0024* (0.0006)
Observations	18,943	20,519	20,204	17,909	14,367	10,350

Primary sample:

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
First-stage F	43.93	131.31	102.39	43.72	52.90	143.70
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Four-wave (EM) sample:

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
First-stage F	29.40	32.91	51.24	43.49	53.01	83.21
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Appendix: Decomposition Ladder — 4 Rungs

The four-wave specification differs from the primary in three ways. We isolate each by running the IV on four nested samples:

	Waves	Sample	Province	Isolates
(i)	2013/18	Full	2011 res.	primary
(ii)	2013/18	Ever-married	2011 res.	EM-restriction (i)→(ii)
(iii)	2013/18	Ever-married	Childhood	Childhood-prov. ME (ii)→(iii)
(iv)	2003/08/13/18	Ever-married	Childhood	Added-waves (iii)→(iv)

Tables on the next four slides — fertility outcomes only.

Appendix: Decomposition Rung (i) — Primary, raw

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.0095 (0.1273)	0.1226 (0.5231)	-0.0431 (1.9677)	0.9235 (1.3419)	-1.5150 (1.7893)	0.1839 (1.3711)
Parity ≥ 2	0.0285 (0.0409)	0.3048 (0.5464)	-0.1508 (0.7457)	0.6126 (1.9059)	1.7403 (3.4194)	-0.8450 (1.1891)
Parity ≥ 3	0.0084 (0.0055)	0.3816 (0.2481)	-0.3712 (0.5649)	1.2502 (1.5588)	2.8929 (2.6999)	1.0620 (1.8097)
Parity ≥ 4	— (—)	— (—)	0.2496 (0.3399)	1.9047** (0.8094)	2.2458 (1.4146)	4.2520** (1.9632)
Parity ≥ 5	— (—)	— (—)	0.3133 (0.2324)	1.3921* (0.8025)	1.4136 (0.9119)	4.1326** (1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Appendix: Decomposition Rung (ii)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.4099 (1.7248)	-0.4872 (1.0891)	0.9206 (2.6042)	-0.8402 (1.2393)	-2.4637 (1.5712)	0.2145 (0.6479)
Parity ≥ 2	0.1292 (0.1952)	1.4537 (2.1300)	1.2401 (1.3208)	-0.5771 (1.9762)	1.6371 (3.3087)	-0.8876 (1.4882)
Parity ≥ 3	0.0359 (0.0922)	1.2429 (0.9420)	-0.1566 (0.8372)	0.7115 (1.6160)	2.9164 (2.7133)	1.3860 (2.2519)
Parity ≥ 4	- (-)	- (-)	-0.1431 (0.4861)	1.7677** (0.8790)	2.2596 (1.3754)	4.6098* (2.3812)
Parity ≥ 5	- (-)	- (-)	0.2610 (0.3487)	1.3126 (0.8506)	1.4374 (0.9243)	4.3275** (1.9093)
Observations	10,721	11,572	11,283	9,746	7,931	5,907

Appendix: Decomposition Rung (iii)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.5138 (2.0057)	-0.4345 (1.0637)	1.4650 (2.7451)	0.3480 (0.7849)	-2.7761 (1.8694)	-0.2932 (0.7883)
Parity ≥ 2	-0.0428 (0.1973)	1.2401 (2.2961)	2.3784* (1.3101)	-0.5379 (1.6543)	0.1194 (2.4396)	-1.3289 (1.4551)
Parity ≥ 3	0.0144 (0.0948)	1.2056 (0.8786)	-0.0581 (0.7228)	-0.3178 (1.5141)	-0.2215 (2.0330)	-1.7220 (2.4169)
Parity ≥ 4	– (–)	– (–)	-0.2682 (0.3732)	1.7497* (0.9399)	2.0977 (1.5434)	0.8757 (2.6843)
Parity ≥ 5	– (–)	– (–)	0.0946 (0.3453)	0.7475 (0.7486)	1.1589 (1.2384)	2.3016* (1.2486)
Observations	10,721	11,572	11,283	9,746	7,931	5,907

Appendix: Decomposition Rung (iv) — 4-Wave

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.6658 (2.2741)	-1.0503 (1.1459)	1.4667 (2.3698)	0.2834 (0.6135)	-2.4309 (1.9752)	0.9380 (0.6439)
Parity ≥ 2	0.1760 (0.1972)	1.0901 (2.3668)	1.9913 (1.3790)	-1.0349 (1.4748)	0.7577 (2.2178)	0.2081 (1.2591)
Parity ≥ 3	0.1306 (0.1237)	1.3511 (0.9421)	-0.5372 (0.8202)	-1.0839 (1.6311)	-0.0772 (2.0124)	-0.9134 (1.9287)
Parity ≥ 4	— (—)	— (—)	-0.3500 (0.3757)	1.2114 (1.0065)	1.1983 (1.5172)	0.6504 (1.8778)
Parity ≥ 5	— (—)	— (—)	0.2083 (0.2610)	1.2281 (0.9975)	1.0989 (1.0670)	0.6201 (1.2463)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Appendix: Region 1–4 ("Other"), Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2609 (0.1868)	-1.3578 (1.0728)	1.7261 (1.7837)	0.0472 (1.2570)	-2.0559 (1.6061)	-1.9054 (0.7543)
Parity ≥ 1	0.0452 (0.2257)	-0.3939 (1.0197)	1.8234 (1.4001)	1.3042 (2.0493)	-6.0697 (2.5172)	-0.9893 (0.9780)
Parity ≥ 2	-0.0363 (0.0429)	-0.5946 (0.6160)	-1.2016 (0.9670)	2.6652 (2.1099)	-4.8945 (3.1711)	-1.1427 (1.6412)
Parity ≥ 3	-0.0006 (0.0014)	0.1853 (0.3480)	-0.6541 (0.7896)	-0.1352 (3.9169)	0.1920 (3.8745)	0.5938 (2.6857)
Parity ≥ 4	– (–)	– (–)	0.2275 (0.4780)	1.4477 (1.3774)	1.4941 (2.3666)	6.1002 (2.5449)
Parity ≥ 5	– (–)	– (–)	0.5797 (0.2703)	0.8921 (1.3467)	1.4853 (1.6360)	5.4374 (2.7984)
Observations	10,352	9,962	9,219	7,836	6,394	4,799

Appendix: Turkish, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.4330 (0.1970)	-1.2864 (0.7963)	-0.1425 (1.6814)	0.2402 (1.6210)	-1.5190 (1.2953)	-2.3851 (2.9491)
Parity ≥ 1	-0.1718 (0.1202)	-0.8128 (0.7729)	-0.5693 (2.3789)	1.2331 (1.9282)	-3.9264 (2.7780)	-2.1423 (3.3247)
Parity ≥ 2	-0.0034 (0.0239)	0.0618 (0.7466)	-1.2228 (0.9056)	0.1915 (2.7639)	-0.7790 (5.4613)	-3.6633 (2.3983)
Parity ≥ 3	0.0041 (0.0043)	0.1653 (0.3170)	-0.3431 (0.5697)	-1.0986 (1.7848)	2.7682 (4.0056)	-0.8493 (3.3966)
Parity ≥ 4	— (—)	— (—)	-0.0748 (0.3250)	-0.3461 (1.0642)	2.5819 (1.0607)	1.8207 (2.9586)
Parity ≥ 5	— (—)	— (—)	0.2606 (0.1146)	-0.0430 (0.3103)	0.0264 (1.5401)	5.1158 (3.8226)
Observations	11,013	10,512	9,655	8,137	6,634	4,960

Appendix: Low Education, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.1848 (0.3031)	-0.8722 (1.2854)	2.0480 (1.8239)	-0.0590 (1.5294)	0.1893 (1.0346)	-0.2025 (1.1901)
Parity ≥ 1	-0.0234 (0.1870)	0.0484 (1.2489)	0.2907 (2.4427)	1.3034 (1.9572)	0.0446 (1.2737)	0.5583 (1.3313)
Parity ≥ 2	0.0029 (0.0571)	0.2609 (1.6357)	-1.1164 (1.6824)	1.6909 (1.6453)	1.6018 (2.7430)	1.9856 (1.6354)
Parity ≥ 3	0.0038 (0.0069)	0.7375 (0.7647)	-0.6375 (1.2099)	2.8227 (2.2384)	3.2607 (2.4464)	4.6894 (1.6967)
Parity ≥ 4	– (–)	– (–)	-0.0667 (0.6786)	3.1747 (1.2937)	2.9678 (1.0527)	6.6579 (2.3695)
Parity ≥ 5	– (–)	– (–)	0.3507 (0.4137)	1.7869 (1.2837)	1.8934 (1.1604)	5.6194 (1.8604)
Observations	9,361	9,132	8,729	7,633	6,348	4,897

Appendix: High Education, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0176 (0.0534)	0.2143 (0.5891)	-0.8859 (3.0142)	-1.4122 (3.2686)	-1.6326 (2.9202)	1.9265 (4.8016)
Parity ≥ 1	0.0414 (0.0289)	-0.3907 (0.6715)	-0.5787 (2.9107)	0.4812 (3.7083)	-9.4252 (4.9289)	0.6246 (5.4977)
Parity ≥ 2	0.0006 (0.0235)	0.3035 (0.2388)	0.2737 (1.1549)	0.1005 (4.6549)	-4.5550 (7.5530)	-15.0970 (8.5615)
Parity ≥ 3	nan (nan)	0.0801 (0.0614)	0.0441 (0.8062)	-3.2287 (1.2656)	7.3840 (6.9828)	-18.7862 (7.4317)
Parity ≥ 4	- (-)	- (-)	-0.5710 (0.4974)	-0.3215 (0.3705)	3.9271 (3.1890)	-8.8020 (4.8629)
Parity ≥ 5	- (-)	- (-)	-0.4857 (0.4814)	0.1304 (0.1855)	-0.3653 (0.5670)	-2.3138 (2.1605)
Observations	4,993	4,352	3,512	2,607	1,897	1,213

Appendix: Childhood Village, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.1211 (0.2736)	-1.4415 (1.1784)	0.8995 (1.5356)	1.0317 (1.4479)	1.0009 (0.9845)	3.4578*** (0.6143)
Parity ≥ 1	0.1742 (0.1738)	-0.1255 (1.2916)	0.4177 (2.2922)	2.6297 (2.4481)	0.5035 (1.5035)	3.3643 (0.8700)
Parity ≥ 2	0.1081 (0.0906)	0.7126 (0.9498)	-1.9560 (1.8797)	3.8106 (2.8029)	-1.0797 (4.0945)	4.7298 (1.4622)
Parity ≥ 3	0.0114 (0.0137)	1.1294 (0.6116)	-1.8534 (2.3677)	4.0855 (4.0760)	1.8258 (3.2759)	4.4503 (2.7838)
Parity ≥ 4	- (-)	- (-)	-0.8554 (1.2278)	2.3753 (2.9829)	10.8030 (5.2495)	4.8970 (2.8621)
Parity ≥ 5	- (-)	- (-)	-0.0620 (0.5388)	-0.5112 (1.3538)	3.3327 (2.6586)	6.2334 (2.0937)
Observations	5,951	5,876	5,551	4,810	4,014	3,144

Appendix: Childhood Province/District Center, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2411 (0.2742)	-0.3869 (1.0264)	0.4872 (2.2987)	-1.6019 (1.3923)	-0.3382 (1.5140)	-5.7737 (2.9858)
Parity ≥ 1	-0.1708 (0.1566)	0.0695 (0.7220)	0.2356 (2.4566)	-1.1591 (1.3131)	-3.1994 (2.1961)	-4.2549 (3.2637)
Parity ≥ 2	-0.0267 (0.0362)	0.1538 (0.5424)	0.4950 (1.2147)	-2.2219 (2.4383)	2.3199 (3.2030)	-6.5088 (2.0111)
Parity ≥ 3	-0.0016 (0.0068)	-0.0079 (0.1930)	0.4753 (1.1148)	-0.4963 (1.6198)	4.8564 (2.6472)	-4.0918 (3.4030)
Parity ≥ 4	— (—)	— (—)	0.6621 (0.4222)	1.4791 (1.1368)	-0.4328 (1.6984)	3.4158 (5.7876)
Parity ≥ 5	— (—)	— (—)	0.4354 (0.2074)	1.9743 (1.0362)	1.7589 (1.1231)	3.3016 (3.5413)
Observations	8,403	7,608	6,690	5,430	4,231	2,966

Appendix: Heterogeneity — Region Heavy, EM (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	3.4637 (1.6930)	-0.1417 (1.0351)	-0.7001 (0.9967)	1.1050 (1.8278)	1.2919 (1.1700)	0.7813 (0.9819)
Parity ≥ 2	0.7230 (0.3124)	4.9204 (2.2036)	3.9068 (2.0116)	-0.0815 (1.8604)	6.4816 (2.1972)	0.7833 (1.0608)
Parity ≥ 3	0.2263 (0.1818)	1.1774 (1.3025)	-0.2640 (1.3782)	4.8317 (2.0548)	9.5649* (2.4606)	4.2373 (1.6056)
Parity ≥ 4	— (—)	— (—)	0.5677 (1.0679)	3.1868 (1.4297)	7.3546 (1.8899)	8.3427 (2.3085)
Parity ≥ 5	— (—)	— (—)	0.3315 (0.5873)	2.4357 (1.1019)	4.3646 (1.6014)	7.3215 (3.8179)
Observations	5,666	6,103	5,823	4,995	3,954	2,796

Appendix: Heterogeneity — Kurdish/Arabic, EM (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	1.4726 (1.2192)	-1.0009 (2.0293)	-1.1131 (2.6882)	-2.2038 (2.1287)	1.7139 (1.8921)	0.3218 (1.4880)
Parity ≥ 2	0.2515 (0.4117)	2.8340 (1.7714)	2.1400 (3.6033)	0.8100 (2.4592)	2.9979 (2.4310)	2.0974 (1.5816)
Parity ≥ 3	-0.0823 (0.3290)	2.0225 (1.9931)	-1.3910 (1.8745)	12.3488** (2.4731)	3.2532 (3.0358)	1.3143 (2.2161)
Parity ≥ 4	— (—)	— (—)	-0.1841 (1.7918)	8.4375 (2.4696)	2.0872 (3.8543)	7.1089 (1.7496)
Parity ≥ 5	— (—)	— (—)	0.3056 (0.9846)	4.1805 (2.6576)	6.1599 (2.8704)	6.0199 (2.0907)
Observations	4,492	4,765	4,517	3,885	3,041	2,157

Appendix: Ideal Fertility — Region 5 and Kurdish/Arabic (RW)

Region 5 (heavy) — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0688 (0.1169)	0.2123 (0.2372)	0.7273* (0.3361)	0.3676** (0.1428)
Observations	4,003	4,003	4,003	4,003

Kurdish / Arabic — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.1287 (0.1578)	0.5124 (0.4172)	0.9109** (0.3501)	0.4754 (0.1844)
Observations	3,177	3,177	3,177	3,177

Appendix: Ideal Fertility — Education and Childhood Residence (RW)

Low education — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0012 (0.0879)	0.2032 (0.2468)	0.6757** (0.2471)	0.3513** (0.1005)
Observations	11,723	11,723	11,723	11,723

Childhood village — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	-0.1961 (0.1081)	-0.0944 (0.2524)	0.5148 (0.4037)	0.2867 (0.2053)
Observations	7,275	7,275	7,275	7,275